

Exposé X — Euler–Poincaré Formula in Étale Cohomology

EULER–POINCARÉ FORMULA IN ÉTALE COHOMOLOGY

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(written up by I. Bucur)

The purpose of the present exposé is to state and prove formula (7.2), of Euler–Poincaré type. This formula generalizes a formula obtained independently by Ogg and Chafarevitch, whose results are presented, from the viewpoint of étale cohomology, in M. Raynaud’s exposé [1].

The Euler–Poincaré formula uses in an essential way the formalism of derived categories. As in loc. cit., the proof is made through a variant (5.1) of a well-known formula of Weil.

1. Sheaves on a scheme with operators

Let X be a scheme endowed with a group G acting by automorphisms. Proceeding as in ([1], chap. V), one may define the notion of an étale sheaf on X with group of operators G , or simply a “sheaf on $(X_{\text{ét}}, G)$ ”.

By definition, to give an étale sheaf on X with group of operators G is to give a sheaf $F \in \text{ob}(\tilde{X}_{\text{ét}})$ and a morphism

$$S_g : F \longrightarrow g^*(F)$$

for each $g \in G$, in such a way that the following conditions are satisfied:

$$\text{a) } S_e = 1_F,$$

and, for $g, h \in G$, the diagram

$$\begin{array}{ccc} F & \xrightarrow{S_{hg}} & (hg)^*(F) \\ S_h \downarrow & & \downarrow C_{h,g}(F) \\ h^*(F) & \xrightarrow{h^*(S_g)} & h^*(g^*(F)) \end{array}$$

is commutative, where

$$C_{h,g}(F) : (hg)^*(F) \longrightarrow h^*(g^*(F))$$

is the canonical isomorphism.

Morphisms between two sheaves (F, S_g) , (F', S'_g) on X with group of operators G are defined in the evident way, and one obtains a category $\tilde{X}_{\text{ét},G}$ which is a topos, as follows readily from Giraud’s criterion (SGA 4 II 4.12).

Given a ring A of $\tilde{X}_{\text{ét},G}$, one may consider the abelian category of A -modules, which will be denoted $\text{Mod}(A)$.

For example, given a ring Λ , one may equip the constant sheaf Λ_X associated with Λ with the trivial structure of a sheaf with group of operators G , using the canonical transitivity isomorphisms

$$S_g : \Lambda_X \xrightarrow{\sim} g^*(\Lambda_X).$$

All the results proved in [1], chap. V, can be extended to this setting, and we leave the details to the reader.

2. Where one proves that a certain complex of $\Lambda[G]$ -modules is perfect

Recall the following result of SGA 4 XVII.

Lemma 2.1. Let X and Y be two schemes and let $f : X \rightarrow Y$ be a proper morphism, with Y quasi-compact. Let A be a torsion ring, and let $F^\bullet$ be a complex of sheaves of A -modules on X of finite Tor-dimension (SGA 4 XVII). Then $Rf_*(F^\bullet)$ is of finite Tor-dimension.

Proposition 2.2. Let X be a scheme proper over a separably closed field k , let G be a finite group acting on X , and let V be an open subset of X stable under G and such that G acts admissibly (SGA 1 V 1.7) and freely on V . Let $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$, with ℓ a prime number different from the characteristic of k , and let Λ_V be the constant sheaf on V associated with Λ , endowed with its trivial structure of a sheaf with group of operators G . Let

$$i : V \longrightarrow X$$

be the immersion of V into X , and put $\Lambda_{V,X} = i_!(\Lambda_V)$. Under these conditions the complex of $\Lambda[G]$ -modules

$$R\Gamma_X(\Lambda_{V,X})$$

is a perfect complex (VIII 7).

Proof. If $p : X \rightarrow Y = X/G$ is the canonical projection, it is well known that there is a canonical isomorphism

$$R\Gamma_X(\Lambda_{V,X}) \simeq R\Gamma_Y(Rp_*(\Lambda_{V,X}))$$

in the category $D(\Lambda[G])$. On the other hand, the complex $Rp_*(\Lambda_{V,X})$ of $\Lambda[G]$ -modules on Y is of finite Tor-dimension. Indeed, since $p : X \rightarrow Y$ is a finite morphism of schemes, one has

$$R^q p_*(\Lambda_{V,X}) = 0 \quad (q \neq 0)$$

(SGA 4 VIII 5.5). Hence

$$Rp_*(\Lambda_{V,X}) = p_*(\Lambda_{V,X}).$$

SGA 4 VIII 5.5 also gives the possibility of determining the fibres of the sheaf $p_*(\Lambda_{V,X})$, and, taking account of the fact that G acts freely on V , one finds

$$(p_*(\Lambda_{V,X}))_{\bar{y}} \simeq \begin{cases} \prod_{x \in p^{-1}(y)} \Lambda_{V,X,\bar{x}} \simeq \Lambda[G], & y \in p(V), \\ 0, & y \notin p(V), \end{cases}$$

where $\bar{y}$ is the geometric point of Y localized at y .

Consequently $p_*(\Lambda_{V,X})$ is flat over $\Lambda[G]$, fibre by fibre (SGA 4 XVII), and therefore is of finite Tor-dimension. Applying 2.1 to the canonical morphism $Y \rightarrow \text{Spec}(k)$, we deduce that

$$R\Gamma_X(\Lambda_{V,X}) = R\Gamma_Y(Rp_*(\Lambda_{V,X}))$$

is of finite Tor-dimension.

To conclude that $R\Gamma_X(\Lambda_{V,X})$ is perfect, it remains to show that it is pseudo-coherent and has bounded cohomology (VIII 8.1). Since $\Lambda[G]$ is noetherian, to verify that $R\Gamma_X(\Lambda_{V,X})$ is pseudo-coherent it is enough to show that the groups $H^q(X, \Lambda_{V,X})$ are of finite type over $\Lambda[G]$ (VIII 7.2), or equivalently over Λ ; this follows from (SGA 4 XIV 1.1), applied to the Λ -module $\Lambda_{V,X}$.

3. Reminders on linear representations of finite groups

In this number, Λ denotes a commutative ring, and G a group.

3.1. Characters

Consider the full subcategory $\mathcal{C}$ of $\text{Mod}(\Lambda[G])$ consisting of $\Lambda[G]$ -modules M whose underlying Λ -module is projective of finite type. For such a module M and for $g \in G$, consider the endomorphism, for the Λ -module structure,

$$\varphi_M : M \longrightarrow M, \quad \varphi_M(x) = gx.$$

One obtains a function

$$\chi_M : G \longrightarrow \Lambda, \quad \chi_M(g) = \text{Tr}_M^*(\varphi_g),$$

which is a central function on G and is called the character of the $\Lambda[G]$ -module M .

Associating to each $M \in \text{ob}(\mathcal{C})$ its character χ_M gives an additive function (VIII 1). Using the formalism of traces for endomorphisms of perfect complexes, one even obtains an additive function Tr on the category of complexes of $\Lambda[G]$ -modules which are perfect as complexes of Λ -modules. Hence an homomorphism

$$\text{Tr} : R_\Lambda^*(G) \longrightarrow \mathcal{F}(G, \Lambda),$$

where $R_\Lambda^*(G)$ is the ring of representations of G over Λ (VIII 10), and $\mathcal{F}(G, \Lambda)$ is the module of central functions $G \rightarrow \Lambda$.

When G is finite, the composite of the canonical homomorphism

$$K^*(\Lambda[G]) \longrightarrow R_\Lambda^*(G)$$

with Tr will be denoted by tr .

Recall the following result (IX th. 1, cor. 3).

Proposition 3.2. Let $\mathbb{Z}_\ell$ be the ring of ℓ -adic integers and let G be a finite group. Then the homomorphism

$$\text{tr} : K^*(\mathbb{Z}_\ell[G]) \longrightarrow \mathcal{F}(G, \mathbb{Z}_\ell)$$

is a monomorphism. Two projective $\mathbb{Z}_\ell[G]$ -modules having the same trace function in $\mathcal{F}(G, \mathbb{Z}_\ell)$ are isomorphic.

3.3. Restriction and induction

Let $\alpha : G \rightarrow G'$ be a homomorphism from a group G to a group G' . By restriction of operators one obtains an exact functor

$$J : \text{Mod}(\Lambda[G']) \longrightarrow \text{Mod}(\Lambda[G]).$$

This functor has a left adjoint

$$I : \text{Mod}(\Lambda[G]) \longrightarrow \text{Mod}(\Lambda[G']),$$

namely

$$I(F) = \Lambda[G'] \otimes_{\Lambda[G]} F.$$

If F is projective of finite type, respectively of finite type, then so is $I(F)$.

If $\varphi \in \mathcal{F}(G', \Lambda)$ is a central function on G' , the function $\varphi \circ \alpha$ is a central function on G , and therefore one obtains an application

$$\alpha^* : \mathcal{F}(G', \Lambda) \longrightarrow \mathcal{F}(G, \Lambda).$$

If φ is the character of a $\Lambda[G']$ -module M in $\mathcal{C}(G')$, then $\alpha^*(\varphi)$ is the character of the $\Lambda[G]$ -module $J(M)$.

Now suppose that $\alpha : G \rightarrow G'$ is injective and that G has finite index in G' . If F is a $\Lambda[G]$ -module in $\mathcal{C}(G)$ whose character is $\psi : G \rightarrow \Lambda$, then $I(F)$ lies in $\mathcal{C}(G')$ and the character $\psi^* : G' \rightarrow \Lambda$ of $I(F)$ is the character induced by ψ , i.e. the function on G' deduced from ψ by the formula

$$\psi^*(g') = \sum_{s \in G'/G} \psi(s^{-1}g's), \quad (3.3.1)$$

with the convention that $\psi(s^{-1}g's) = 0$ if $s^{-1}g's \notin G$.

3.4. Additional structures on complexes

Let $X^\bullet$ and $Y^\bullet$ be two complexes of Λ -modules. As in the case of modules, if one of the complexes $X^\bullet, Y^\bullet$ has an additional structure of a Λ - B -bimodule, it is possible to introduce on $\text{Hom}_\Lambda(X^\bullet, Y^\bullet)$ a structure of complex of B -modules, and consequently a corresponding structure on $\text{RHom}_\Lambda(X^\bullet, Y^\bullet)$.

For example, suppose that $X^\bullet$ is an object of $D^-(\Lambda)$ and that $Y^\bullet$ is a complex of Λ - B -bimodules. In particular, $Y^\bullet$ is an object of $D(\Lambda)$. Let $P^\bullet \rightarrow X^\bullet$ be a projective resolution of $X^\bullet$. Then $\text{Hom}_\Lambda(P^\bullet, Y^\bullet)$ is evidently a complex of left B -modules, i.e. an object of $D(B)$. If another projective resolution $P_1^\bullet \rightarrow X^\bullet$ is used, the complex of B -modules $\text{Hom}_\Lambda(P_1^\bullet, Y^\bullet)$ is canonically isomorphic in $D(B)$ to $\text{Hom}_\Lambda(P^\bullet, Y^\bullet)$. It follows that one may define $\text{RHom}_\Lambda(X^\bullet, Y^\bullet)$ as an object of $D(B)$, determined up to unique isomorphism, and functorial in $X^\bullet$ in $D^-(\Lambda)$ and in $Y^\bullet$ in $D(\Lambda, B)$.

Analogous considerations apply to the complex $X^\bullet \otimes_\Lambda^L Y^\bullet$. For example, if $Y^\bullet$ is a complex of Λ - B -bimodules as above and $X^\bullet$ is a right bounded complex of right Λ -modules, one may define $X^\bullet \otimes_\Lambda^L Y^\bullet$ as an object of $D(B)$ determined up to unique isomorphism.

3.5

Let Λ be a commutative ring, let A and B be two algebras over Λ , and put $C = A \otimes_\Lambda B$. Thus C -modules are A - B -bimodules.

If $\mathcal{A}$ is a full triangulated subcategory of $D(\Lambda)$, denote by $D(C)_\mathcal{A}$ the full triangulated subcategory of $D(C)$ generated by the complexes which, when regarded as objects of $D(\Lambda)$, belong to $\mathcal{A}$.

Proposition 3.6. With the notation of 3.4 and 3.5, let $\mathcal{A}$ be a full triangulated subcategory of $D(\Lambda)$ stable under direct factors. If $P^\bullet$ is an object of $D(B)_{\text{parf}}$ (VIII 7) and $M^\bullet$ is an object of

$D(C)_{\mathcal{A}}$, then the complex of Λ -modules $\mathrm{RHom}_B(P^\bullet, M^\bullet) \in D(\Lambda)$ defined in 3.4 is an object of $\mathcal{A}$. In particular, there is a pairing of triangulated categories

$$\mathrm{RHom}_B : D(B)_{\mathrm{parf}} \times D(C)_{\mathcal{A}} \longrightarrow \mathcal{A}, \quad (3.6.1)$$

that is, a bifunctor exact in each variable, and consequently (VIII 3) a bilinear map

$$K^*(B) \times K^*(D(C)_{\mathcal{A}}) \longrightarrow K^*(\mathcal{A}). \quad (3.6.2)$$

3.7

Suppose in particular that $B = \Lambda[G]$ and $C = A[G] = A \otimes_{\Lambda} \Lambda[G]$. Given a complex $X^\bullet$ of $A[G]$ -modules, denote by $X^{\bullet G}$ the complex of A -modules which is the image of $X^\bullet$ under the functor that associates to each $A[G]$ -module X the A -module of elements of X invariant under G .

Proposition 3.8. Let G be a finite group, let $P^\bullet$ be a bounded complex of $A[G]$ -modules all of whose components are projective of finite type, and let $M^\bullet$ be a complex of $A[G]$ -modules. Under these conditions there are canonical isomorphisms of complexes of A -modules

$$\mathrm{Hom}_{A[G]}(P^\bullet, M^\bullet) = \mathrm{Hom}_A(P^\bullet, M^\bullet)^G = (P^{\bullet*} \otimes_A M^\bullet)^G = P^{\bullet*} \otimes_{A[G]} M^\bullet. \quad (3.8.1)$$

The proof uses the well-known isomorphisms in the case of modules and is left to the reader. Note that, in the second and third terms, the operations of G on $\mathrm{Hom}_A(P^\bullet, M^\bullet)$, respectively on $P^{\bullet*} \otimes_A M^\bullet$, are defined by transport of structure from its operations on $P^\bullet$ and $M^\bullet$, respectively $P^{\bullet*}$ and $M^\bullet$.

Corollary 3.9. If $P^\bullet \in \mathrm{ob}(D(A[G])_{\mathrm{parf}})$ and $M^\bullet \in D(A[G])$, then in the derived category $D(A)$ one has the isomorphism

$$\mathrm{RHom}_{A[G]}(P^\bullet, M^\bullet) = \mathrm{RHom}_A(P^\bullet, M^\bullet)^G. \quad (3.9.1)$$

Indeed, since $P^\bullet \in D(A[G])_{\mathrm{parf}}$, we may suppose that $P^\bullet$ is bounded and that all its components are projective of finite type. Hence

$$\mathrm{RHom}_{A[G]}(P^\bullet, M^\bullet) = \mathrm{Hom}_{A[G]}(P^\bullet, M^\bullet),$$

and one can use the isomorphisms (3.8.1). But $P^{\bullet*}$ is, like $P^\bullet$, bounded with projective finite-type components, hence

$$P^{\bullet*} \otimes_A^L M^\bullet = P^{\bullet*} \otimes_A M^\bullet.$$

Similarly,

$$P^{\bullet*} \otimes_A^L M^\bullet = P^{\bullet*} \otimes_A M^\bullet.$$

Moreover the components of the complex $P^{\bullet*} \otimes_A M^\bullet$ are cohomologically trivial ([3] IX 3), and therefore

$$R\Gamma^G(P^{\bullet*} \otimes_A^L M^\bullet) = (P^{\bullet*} \otimes_A M^\bullet)^G.$$

4. The Swan representation

Let E be a smooth connected algebraic curve over the algebraically closed field k . If $x \in E$ is a closed point of E , its local ring $\mathcal{O}_{E,x}$, or simply $\mathcal{O}_x$, is a discrete valuation ring whose field of fractions is isomorphic to the field of rational functions on E and whose residue field is k . The corresponding valuation will be denoted v_x .

Lemma 4.1. Let E be a smooth algebraic curve over the algebraically closed field k , and let $g : E \rightarrow E$ be an endomorphism of E different from the identity. Suppose that $x \in E$ is an isolated fixed point of g , and let π be a uniformizer of $\mathcal{O}_{E,x}$. If v_x is the corresponding valuation of $\mathcal{O}_{E,x}$, then the multiplicity $v_x(g)$ of the fixed point x of the map g (IV) is equal to $v_x(g(\pi) - \pi)$.